Final answers (record here — graded on the answer; show your work):

1. ________ 2. ________ 3. ________ 4. ________ 5. ________ 6. ________ 7. ________
8. ________ 9. ________ 10. ________ 11. ________ 12. ________

Function Families

Example (Name the parent & give the domain):
$f(x) = 2^x$ is the **exponential** parent: it passes through $(0, 1)$, never touches the x-axis, and its domain is all real numbers.

Directions: Name the parent family and state the domain.

1. $f(x) = x^2$

2. $f(x) = x^3$

3. $f(x) = \sqrt{x}$

4. $f(x) = \dfrac{1}{x}$

5. $f(x) = |x|$

6. $f(x) = x$

7. $f(x) = 2^x$

8. $f(x) = \sqrt[3]{x}$

9. $f(x) = \dfrac{1}{x^2}$

10. $f(x) = 5$

11. $f(x) = x^4$

Challenge (same sheet):

12. $f(x) = \log_2 x$ — name the family, give the domain, and the one point every log parent passes through.

Answer Key — fully worked

Procedure: Match the rule to a known family; Recall the family's basic shape and a key point; State the domain — watch square roots and denominators.

1. Quadratic; domain all reals $(-\infty, \infty)$
2. Cubic; domain all reals
3. Square root; domain $x \geq 0$
4. Reciprocal (rational); domain $x \neq 0$
5. Absolute value; domain all reals
6. Linear (identity); domain all reals
7. Exponential; domain all reals
8. Cube root; domain all reals
9. Rational; domain $x \neq 0$
10. Constant; domain all reals
11. Quartic (even power); domain all reals
12. Logarithmic; domain $x > 0$; passes through $(1, 0)$

Final answers (record here — graded on the answer; show your work):

1. ________ 2. ________ 3. ________ 4. ________ 5. ________ 6. ________ 7. ________
8. ________ 9. ________ 10. ________ 11. ________

Transformations

Example (Describe the transformation & give the new vertex):
$g(x) = (x+4)^2 - 1$: inside $+4$ shifts **left 4**, outside -1 shifts **down 1**, so the vertex is $(-4, -1)$.

Directions: Describe each transformation from the parent x^2 and give the new vertex.

1. $y = (x-2)^2$

2. $y = x^2 + 3$

3. $y = (x+5)^2 - 2$

4. $y = -(x-1)^2$

5. $y = (x-3)^2 + 4$

6. $y = x^2 - 6$

7. $y = (x+2)^2$

8. $y = -(x+4)^2 + 1$

9. $y = (x-1)^2 - 5$

10. $y = -x^2 + 2$

Challenge (same sheet):

11. $y = -(x+2)^2 + 5$ — describe every transformation, give the vertex, and state whether it opens up or down.

Answer Key — fully worked

Procedure: Read the inside $(x-h)$: shift opposite the sign (left/right); Read the outside $+k$: shift up/down; A leading $-$ reflects over the x-axis; Vertex lands at (h,k).

1. Right 2; vertex $(2,0)$

2. Up 3; vertex $(0,3)$

3. Left 5, down 2; vertex $(-5,-2)$

4. Right 1, reflect over x-axis; vertex $(1,0)$, opens down

5. Right 3, up 4; vertex $(3,4)$

6. Down 6; vertex $(0,-6)$

7. Left 2; vertex $(-2,0)$

8. Left 4, up 1, reflect over x-axis; vertex $(-4,1)$, opens down

9. Right 1, down 5; vertex $(1,-5)$

10. Up 2, reflect over x-axis; vertex $(0,2)$, opens down

11. Left 2, up 5, reflect over x-axis; vertex $(-2,5)$, opens down

Final answers (record here — graded on the answer; show your work):

1. ________ 2. ________ 3. ________ 4. ________ 5. ________ 6. ________ 7. ________
8. ________ 9. ________ 10. ________ 11. ________

Domain & Range

Example (State the domain (and range where asked)):
$f(x) = \dfrac{1}{x+3}$: the denominator can't be 0, so $x \neq -3$. Domain: all reals except -3.

Directions: State the domain. For #9 and #10 also state the range.

1. $f(x) = \dfrac{1}{x-4}$

6. $f(x) = \dfrac{2}{x^2-16}$

2. $f(x) = \sqrt{x+6}$

7. $f(x) = \sqrt{2x-8}$

3. $f(x) = \dfrac{x}{x^2-9}$

8. $f(x) = \dfrac{x+1}{x^2-x-6}$

4. $f(x) = \dfrac{1}{x+7}$

9. $f(x) = x^2 - 5$

5. $f(x) = \sqrt{x-1}$

10. $f(x) = |x| + 2$

Challenge (same sheet):

11. $f(x) = \dfrac{\sqrt{x-1}}{x-4}$ — give the domain (both traps apply at once).

Answer Key — fully worked

Procedure: Denominator $\neq 0$ — exclude those x; Even root's inside must be ≥ 0 — solve it; Otherwise the domain is all reals; Range: think about the lowest or highest output.

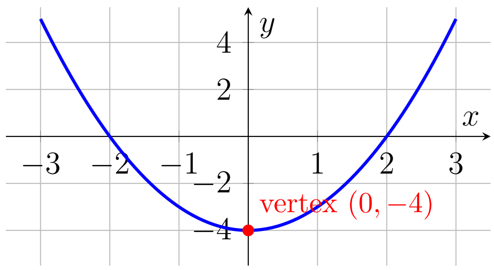
1. $x \neq 4$
2. $x \geq -6$
3. $x \neq \pm 3$ (since $x^2 - 9 = 0$ at ± 3)
4. $x \neq -7$
5. $x \geq 1$
6. $x \neq \pm 4$ (since $x^2 - 16 = 0$ at ± 4)
7. $x \geq 4$ (need $2x - 8 \geq 0$)
8. $x \neq 3,\ -2$ (denominator $(x-3)(x+2) = 0$)
9. Domain all reals; range $y \geq -5$
10. Domain all reals; range $y \geq 2$
11. $x \geq 1$ and $x \neq 4$ (root needs $x \geq 1$; denominator excludes 4)

Final answers (record here — graded on the answer; show your work):

1. ________ 2. ________ 3. ________ 4. ________ 5. ________ 6. ________ 7. ________
8. ________ 9. ________

Quadratic Functions: Vertex, Max & Min

Example (State max or min, the vertex, and the intervals):
For $y = x^2 - 4$ the vertex is $(0, -4)$, a **minimum**. It decreases for $x < 0$ and increases for $x > 0$.



Directions: For each function state max or min, give the vertex, and the increasing / decreasing intervals.

1. $y = x^2 + 2$

4. $y = -(x + 1)^2 - 2$

2. $y = -x^2 + 5$

5. $y = (x + 2)^2 - 1$

3. $y = (x - 3)^2$

6. $y = -(x - 4)^2 + 3$

7. $y = x^2 - 6x$

8. $y = 2(x-1)^2 + 3$

Challenge (same sheet):

9. $y = (x-2)^2 - 9$ — give the vertex, both intervals, and the two x-intercepts.

Answer Key — fully worked

Procedure: Opens up ($+x^2$) gives a minimum; opens down gives a maximum; Vertex form $a(x-h)^2+k$ has vertex (h,k); The graph turns (changes direction) at the vertex.

1. Opens up $\Rightarrow$ min; vertex $(0,2)$; dec $x<0$, inc $x>0$

2. Opens down $\Rightarrow$ max; vertex $(0,5)$; inc $x<0$, dec $x>0$

3. Vertex $(3,0)$, opens up $\Rightarrow$ min; dec $x<3$, inc $x>3$

4. Vertex $(-1,-2)$, opens down $\Rightarrow$ max; inc $x<-1$, dec $x>-1$

5. Vertex $(-2,-1)$, opens up $\Rightarrow$ min; dec $x<-2$, inc $x>-2$

6. Vertex $(4,3)$, opens down $\Rightarrow$ max; inc $x<4$, dec $x>4$

7. Complete the square: $x^2-6x=(x-3)^2-9$; vertex $(3,-9)$ min; dec $x<3$, inc $x>3$

8. $a=2>0$ opens up $\Rightarrow$ min; vertex $(1,3)$; dec $x<1$, inc $x>1$

9. Vertex $(2,-9)$ min; dec $x<2$, inc $x>2$; intercepts $x=-1,\ 5$

Final answers (record here — graded on the answer; show your work):

1. ________ 2. ________ 3. ________ 4. ________ 5. ________ 6. ________ 7. ________
8. ________ 9. ________ 10. ________

Even, Odd, or Neither

Example (Determine even, odd, or neither):
$f(x) = x^4 - x^2$: $f(-x) = x^4 - x^2 = f(x)$, so it is **even**.

Directions: Compute $f(-x)$ and decide: even, odd, or neither. Show the substitution.

1. $f(x) = x^2 + 1$

2. $f(x) = x^3 - x$

3. $f(x) = x^3 + 2$

4. $f(x) = |x|$

5. $f(x) = 2x$

6. $f(x) = x^4 - 3x^2$

7. $f(x) = x^5$

8. $f(x) = x^2 + x$

9. $f(x) = \dfrac{1}{x}$

Challenge (same sheet):

10. $f(x) = x^3 + x^2$ — test $f(-x)$ and classify; explain why a single x^2 term breaks odd symmetry.

Answer Key — fully worked

Procedure: Replace every x with $-x$ and simplify; If $f(-x) = f(x)$ it's even (y-axis symmetry); If $f(-x) = -f(x)$ it's odd (origin symmetry); Otherwise it's neither.

1. $f(-x) = x^2 + 1 = f(x)$: even

2. $f(-x) = -x^3 + x = -(x^3 - x) = -f(x)$: odd

3. $f(-x) = -x^3 + 2$: neither

4. $f(-x) = |-x| = |x| = f(x)$: even

5. $f(-x) = -2x = -f(x)$: odd

6. $f(-x) = x^4 - 3x^2 = f(x)$: even (all even powers)

7. $f(-x) = -x^5 = -f(x)$: odd

8. $f(-x) = x^2 - x$: neither

9. $f(-x) = \dfrac{1}{-x} = -\dfrac{1}{x} = -f(x)$: odd

10. $f(-x) = -x^3 + x^2$: neither — the even x^2 term stays + while x^3 flips, so it can't equal $\pm f(x)$